Lesson 1: Patterns, Sequences and the General Term

Assignment

1. Square tiles are arranged in growing "L" shapes. Figure 1 uses 3 tiles, Figure 2 uses 5 tiles and Figure 3 uses 7 tiles.

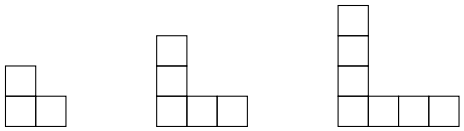


Figure 1 Figure 2 Figure 3 Figure 4

(a) Draw Figure 4 in the space above.

(b) Complete the table.

Figure, n	1	2	3	4	5
Tiles, t_n	3	5	7		

(c) Describe how to get each term from the previous term. _____

(d) Is the sequence arithmetic, geometric or neither? Is it a linear or non-linear function?

2. A sheet of paper is cut in half, then each piece is cut in half again, and so on. After 1 cut there are 2 pieces, after 2 cuts there are 4 pieces.

(a) Write the number of pieces after 1, 2, 3, 4 and 5 cuts. _____

(b) Is the sequence arithmetic, geometric or neither? State d or r. _____

(c) How many pieces are there after 10 cuts?

3. Classify each sequence as arithmetic, geometric or neither. If it is arithmetic, state d; if it is geometric, state r.

(a) $11, 14, 17, 20, \ldots$ (b) $5, 15, 45, 135, \ldots$ (c) $2, 4, 7, 11, \ldots$

(d) $30, 21, 12, 3, \ldots$ (e) $-4, 8, -16, 32, \ldots$ (f) $1, 4, 9, 16, 25$

(g) $200, 100, 50, 25, \ldots$ (h) $-\frac{3}{2}, -1, -\frac{1}{2}, 0, \ldots$ (i) $1.1, 1.21, 1.331, \ldots$

4. State whether each sequence in question 3 is finite or infinite.

5. Write the next two terms of each sequence, and describe the rule using only addition or multiplication.

(a) $9, 16, 23, 30, \ldots$ (b) $81, 27, 9, 3, \ldots$ (c) $-2, 10, -50, \ldots$

(d) $4.5, 3.25, 2, 0.75, \ldots$ (e) $\frac{2}{3}, 2, 6, 18, \ldots$ (f) $1, 1, 2, 3, 5, \ldots$

6. For each general term, write the first four terms.

(a) $t_n = 6n + 1$ (b) $t_n = 20 - 3n$ (c) $t_n = 2(5)^{n-1}$

(d) $t_n = n^2 + 1$ (e) $t_n = 64\left(\frac{1}{2}\right)^n$ (f) $t_n = (-1)^n \cdot n$

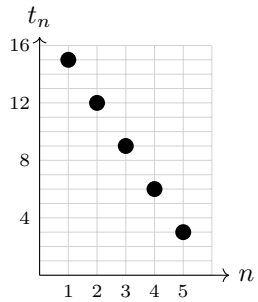
7. A sequence has general term $t_n = 4n + 7$.

(a) Determine t_{25}.

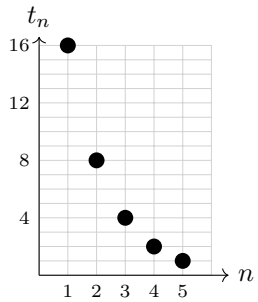
(b) Which term has the value 131?

(c) Explain why 150 is not a term of the sequence.

8. The points (n, t_n) of two sequences are graphed below.



Sequence P



Sequence Q

(a) Which sequence is arithmetic? How can you tell from the graph?

(b) Write the first five terms of each sequence and state d or r.

(c) Determine t_7 for each sequence.

(d) Why are the points not joined with a line or curve?

9. Seats in a theatre: row 1 has 18 seats and each row after that has 2 more seats than the row in front of it. The last row is row 22.

(a) Is the sequence of seats per row finite or infinite? Arithmetic or geometric?

(b) How many seats are in row 5? in row 22?

10. The first term of a sequence is 4 and each term after that is found with the rule "multiply the previous term by 3, then subtract 2."

(a) Write the first five terms.

(b) Is this sequence arithmetic, geometric or neither? Explain.

11. Which sequence is geometric?

A. $3, 6, 9, 12, \ldots$ **B.** $3, 9, 27, 81, \ldots$ **C.** $3, 9, 15, 21, \ldots$ **D.** $3, 4, 6, 9, \ldots$

12. The general term of a sequence is $t_n = 50 - 4n$. The first negative term is

A. t_{12} **B.** t_{13} **C.** t_{14} **D.** t_{50}

13. In the Fibonacci sequence $1, 1, 2, 3, 5, 8, \ldots$, the value of t_{12} is

A. 89 **B.** 144 **C.** 233 **D.** 12

14. Numerical Response A sequence has general term $t_n = 3n^2 - n$. The value of t_{11} is _____.

Lesson 2: Arithmetic Sequences

Assignment

1. Each sequence is arithmetic. State d and the next three terms.

(a) $9, 20, 31, \ldots$

(b) $-14, -6, 2, \ldots$

(c) $55, 38, 21, \ldots$

(d) $2.6, 0.9, -0.8, \ldots$

(e) $\frac{1}{4}, \frac{2}{3}, \frac{13}{12}, \ldots$

(f) $5x - y,\ 3x + 2y,\ x + 5y, \ldots$

2. One term and the common difference of each arithmetic sequence are given. Fill in the missing terms.

(a) $d = 4$

		9		

(b) $d = -6$

			-2	

(c) $d = -1.5$

	0.5			

(d) $d = \frac{1}{2}$

				13

3. Write the first four terms of each arithmetic sequence.

(a) $t_1 = -9$, $d = 5$

(b) $t_3 = 26$, $d = -4$

(c) $t_6 = 40$, $d = 7$

4. Consider the arithmetic sequence $23, 17, 11, 5, \ldots$

(a) Determine the general term in simplest form.

(b) Determine t_{40}.

(c) Which of -169 and -190 are terms of the sequence? For each term, state its position.

5. Determine the general term in simplest form and t_{20} for each arithmetic sequence.

(a) $4, 13, 22, \ldots$

(b) $-30, -18.5, -7, \ldots$

(c) $x + 5y,\ 3x + 4y,\ 5x + 3y, \ldots$

6. Square tiles make a pattern of plus signs. Each arm of Figure n has n tiles.

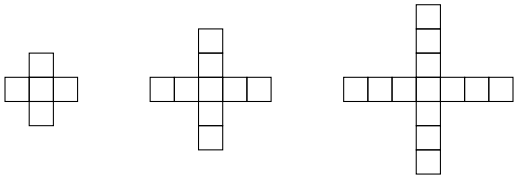


Figure 1 Figure 2 Figure 3

(a) State the number of tiles in Figures 1 to 5. _____

(b) Determine the general term.

(c) How many tiles are in Figure 50?

(d) Which figure uses 145 tiles? Can any figure use exactly 100 tiles?

7. Determine the number of terms in each sequence.

(a) $5, 9, 13, \ldots, 201$ (b) $-17, -25, -33, \ldots, -305$ (c) $2.5, 3.25, 4, \ldots, 22$

8. (a) How many multiples of 8 are there from 40 to 480, inclusive?

(b) State the first and last multiples of 13 between 200 and 900, then find how many multiples of 13 there are between 200 and 900.

9. How many whole numbers between 1 and 1000 leave a remainder of 2 when divided by 7?

10. (a) An arithmetic sequence has $d = -4$ and $t_{18} = -50$. Determine t_1.

(b) An arithmetic sequence has $t_1 = -6$ and $t_{21} = 44$. Determine d and the general term.

(c) An arithmetic sequence starts at 100 and has $d = -3.5$. Its last term is -75. How many terms does it have?

11. Insert the arithmetic means.

(a) four means between -13 and 22 (b) three means between $\frac{1}{2}$ and $\frac{5}{2}$ (c) six means between 3.2 and -1

12. Each set of expressions forms three consecutive terms of an arithmetic sequence. Find the value of the variable and the three terms.
(a) $4x+1,\ \ 3x+6,\ \ x+9$ (b) $2k+1,\ \ 5k-2,\ \ 6k+3$ (then also give the general term)

13. A student says the general term of $3, 7, 11, 15, \ldots$ is $t_n = 3+4n$. Explain the error and write the correct general term.

14. Which sequence is arithmetic with a common difference of -5?
A. $-20, -15, -10, -5, \ldots$
B. $30, 25, 20, 15, \ldots$
C. $125, 25, 5, 1, \ldots$
D. $5, -5, 5, -5, \ldots$

15. The expressions $m+1$, $2m+4$ and $4m-1$, in that order, form an arithmetic sequence. The common difference of the sequence is
A. 8 **B.** 11 **C.** 9 **D.** 20

16. Sequence P is $7, 11, 15, \ldots$ and sequence Q is $200, 194, 188, \ldots$ The first position n at which the term of P is greater than the term of Q is
A. $n = 19$ **B.** $n = 20$ **C.** $n = 21$ **D.** $n = 22$

17. Numerical Response Nine arithmetic means are inserted between -12 and 53. The sixth term of the resulting sequence is _____.

18. Numerical Response The sequence $400, 393, 386, \ldots$ continues forever. The number of positive terms in the sequence is _____.

Lesson 3: Arithmetic Sequences in Context

Assignment

1. For each arithmetic sequence, determine d, t_1 and the general term.
(a) $t_5 = 23,\ t_{12} = 51$ (b) $t_2 = 11,\ t_{15} = -41$ (c) $t_8 = 2,\ t_{20} = 8$

2. An arithmetic sequence has $t_4 = -1$ and $t_{10} = 20$. Use a system of equations to determine t_1 and d, then determine t_{30}.

3. An arithmetic sequence has $t_5 = 31$ and $t_9 = 7$.
(a) Determine t_6, t_7 and t_8 without finding t_1.

(b) Which term of the sequence is -101?

4. An arithmetic sequence has $t_6 = 37$ and $t_{14} = 77$. Is 100 a term of the sequence? Explain.

5. A store stacks cans in a display. Rows 1 and 2 (at the bottom) each have 40 cans, rows 3 and 4 each have 37 cans, rows 5 and 6 each have 34 cans, and so on. The top two rows each have 4 cans.
(a) How many rows are in the display?

(b) How many cans are in row 15? in row 24?

6. Leah deposits \$3200 in an account that pays 4.25% simple interest per year.
(a) How much interest is paid each year? Write the account values at the end of years 1, 2 and 3.

(b) Let t_1 be the value at the end of year 1. Write the general term.

(c) What is the value after 20 years?

(d) After how many full years will the account first hold at least double the deposit?

7. A caterer charges a \$350 set-up fee plus \$28 per guest.

(a) Write the costs for 1, 2, 3 and 4 guests. _____

(b) Write the general term in the form $t_n = t_1 + (n-1)d$, then simplify it to the form $C = mn + b$. What do m and b represent?

(c) Find the cost for 85 guests.

(d) A family's budget is \$3000. What is the greatest number of guests they can invite?

8. An engineer's starting salary is \$48 500 in year 1, with a raise of \$1850 at the start of each year after that.

(a) What is the salary in year 9?

(b) In which year is the salary first more than \$70 000?

9. A delivery van costs \$52 000 new. It loses \$9000 of value in its first year and \$3800 in each year after that.

(a) What is the van worth at the end of year 10?

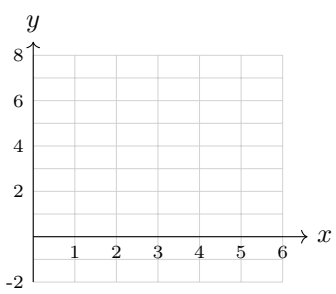
(b) At the end of which year is it first worth less than \$5000?

10. A candle is 24 cm tall when it is lit, and it burns down 1.8 cm every hour. Let $t_1 = 24$.

(a) Write the general term. Which term gives the height after 5 hours, and what is that height?

(b) After how many full hours is the candle first shorter than 3 cm?

11. The linear function $y = 2x - 3$ has its domain restricted to the natural numbers.



(a) Plot the points of the function for $x = 1$ to 5.

(b) List the first five elements of the range. _____

(c) Show that the range values form an arithmetic sequence and state d.

(d) Why are the points not joined?

12. A swimmer swims 400 m on day 1 of a training plan and 150 m more each day after that. The first day she swims more than 5 km is day

A. 30 **B.** 31 **C.** 32 **D.** 33

13. A machine bought for \$18 000 loses \$1500 of value each year. Its value after n years is

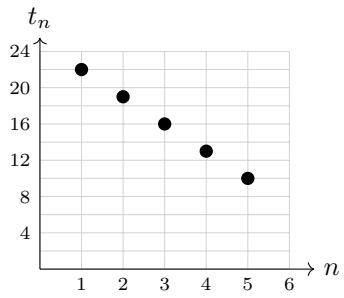
A. $18\,000 - 1500n$

B. $18\,000 - 1500(n - 1)$

C. $16\,500 - 1500n$

D. $18\,000 + 1500n$

Use this graph of an arithmetic sequence for the next two questions.



14. If the pattern in the graph continues, the 20th term of the sequence is

A. -35 **B.** -38 **C.** -32 **D.** -60

15. Numerical Response The general term of the sequence in the graph can be written $t_n = mn + b$. The value of b is _____.

Lesson 4: Arithmetic Series

Assignment

1. Consider the arithmetic sequence $9, 5, 1, -3, -7, \ldots$

(a) Write the series of the first five terms. _____

(b) Complete the table of partial sums.

n	1	2	3	4	5
S_n					

(c) Explain why the partial sums start to decrease.

2. Use pairing (write the sum forwards and backwards) to find each sum.

(a) $1 + 2 + 3 + \cdots + 249 + 250$

(b) $3 + 6 + 9 + \cdots + 147 + 150$

3. Determine the sum of each arithmetic series.

(a) $5 + 12 + 19 + \ldots$ (first 20 terms)

(b) $-15 + (-11) + (-7) + \ldots$ (first 32 terms)

(c) $1.8 + 2.3 + 2.8 + \ldots$ (first 24 terms)

(d) $\frac{1}{3} + \frac{5}{6} + \frac{4}{3} + \ldots$ (first 10 terms)

4. Determine the sum of each arithmetic series from its first and last terms.

(a) $t_1 = 14$, $t_{40} = 170$

(b) $t_1 = -6$, $t_{31} = -126$

5. Determine each sum.

(a) $8 + 17 + 26 + \cdots + 251$

(b) $64 + 58 + 52 + \cdots + (-44)$

6. How many terms of each series must be added to give the stated sum?

(a) $3 + 5 + 7 + \cdots = 195$

(b) $-10 + (-6) + (-2) + \cdots = 320$

7. (a) An arithmetic series has $S_{16} = 432$ and $d = 2$. Determine t_1.

(b) An arithmetic series has 22 terms. The last term is 111 and the sum is 1287. Determine t_1 and d.

8. Seven arithmetic means are placed between -20 and 28. Find the sum of the seven means *without* finding the means.

9. Tamsin saves \$5.00 in the first week of a savings plan. Each week she saves \$1.25 more than the week before, for 30 weeks.

(a) How much does she save in week 30?

(b) How much does she save in total?

10. A gardener waters a straight row of 20 trees. The first tree is 10 m from the tap and the trees are 4 m apart. She fills one bucket at the tap, carries it to the first tree and walks back, then does the same for the second tree, and so on. How far has she walked when she has watered all 20 trees and is back at the tap?

11. The rows of seats in a theatre form an arithmetic sequence. Row 6 has 34 seats, row 15 has 52 seats, and there are 24 rows.

(a) How many seats are in row 1?

(b) How many seats are in the theatre?

12. An arithmetic series has $S_n = 2n^2 - 5n$.

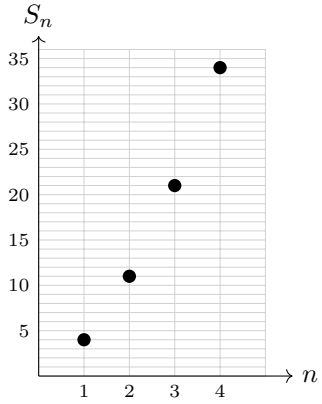
(a) Determine the first four terms of the sequence.

(b) State d. _____

(c) Determine t_{15}.

(d) A different series has $S_n = n^2 + 3n + 1$. Show that its terms do *not* form an arithmetic sequence.

13. The graph shows the points (n, S_n) for an arithmetic series.



(a) Determine t_1, t_2, t_3 and t_4.

(b) State d and determine S_{10}.

14. Which of the following is an arithmetic series?

A. $2 + 6 + 18 + 54$ **B.** $5, 9, 13, 17$ **C.** $20 + 14 + 8 + 2$ **D.** $1 + 4 + 9 + 16$

15. The sum of the first 40 terms of the series $12 + 9 + 6 + \ldots$ is

A. -1920 **B.** -1860 **C.** -3720 **D.** -105

16. An arithmetic series has first term 7, last term 99 and sum 1272. The number of terms is

A. 12 **B.** 23 **C.** 24 **D.** 25

17. Numerical Response The general term of an arithmetic sequence is $t_n = 6n + 5$. The sum of the first 30 terms is _____.

18. Numerical Response The number of terms of the series $15 + 17 + 19 + \ldots$ needed to give a sum of 480 is _____.

Lesson 5: Geometric Sequences

Assignment

1. The first two terms of a sequence are $4, 8, \ldots$ Is this enough information to decide whether the sequence is arithmetic or geometric? Explain.

2. Identify each sequence as arithmetic, geometric or neither. State d or r where there is one, and write the next two terms.

(a) $7, 28, 112, \ldots$

(b) $1000, 200, 40, \ldots$

(c) $3, 5, 9, 15, 23, \ldots$

(d) $-81, 54, -36, \ldots$

(e) $12, 4, -4, \ldots$

(f) $\sqrt{5}, 5, 5\sqrt{5}, \ldots$

(g) $2m, 2m^3, 2m^5, \ldots$

(h) $0.4, 0.04, 0.004, \ldots$

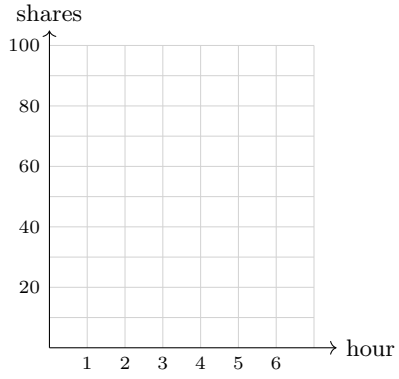
3. A post is shared online. In the first hour 3 people share it, and in each hour after that twice as many people share it as in the hour before.

(a) Write the number of shares in each of hours 1 to 6.
_____

(b) Is the sequence arithmetic or geometric? State d or r.
_____

(c) Plot the points (hour, shares). Is the relationship linear?

(d) Write the general term. _____



4. Write the general term of each geometric sequence.

(a) $6, 24, 96, \ldots$

(b) $80, 20, 5, \ldots$

(c) $-7, 14, -28, \ldots$

(d) $\frac{1}{2}, -\frac{3}{2}, \frac{9}{2}, \ldots$

5. (a) Write the general term of $25, 125, 625, \ldots$ as a single power of 5.

(b) Write the general term of $\frac{1}{9}, \frac{1}{3}, 1, 3, \ldots$ as a single power of 3.

(c) Kai writes the general term of $3, 15, 75, \ldots$ as $t_n = 15^{n-1}$. Explain his error and correct it.

6. Determine the exact value of the indicated term.

(a) $4, 12, 36, \ldots;\ t_{10}$

(b) $160, 80, 40, \ldots;\ t_{12}$

(c) $\frac{2}{3}, -2, 6, \ldots;\ t_9$

(d) $0.5, 1, 2, \ldots;\ t_{15}$

7. Determine the number of terms in each sequence.

(a) $5, 15, 45, \ldots, 10\,935$

(b) $-3, 6, -12, \ldots, -3072$

(c) $4096, 1024, 256, \ldots, \frac{1}{16}$

8. (a) A geometric sequence has 11 terms, $t_1 = 3$ and $r = 2$. Determine the middle term.

(b) Five geometric means are placed between two numbers. How many terms does the sequence have? _____

9. Two terms of a geometric sequence are given. Determine r, t_1 and the general term.

(a) $t_2 = -6,\ t_5 = 162$

(b) $t_4 = 40,\ t_7 = 5$

10. (a) In a geometric sequence $t_3 = 50$ and $t_5 = 1250$. Determine t_1 and the two possible values of t_6.

(b) In a geometric sequence $t_2 = 6$ and $t_6 = \frac{3}{8}$. Explain why there are two sequences and write the first four terms

of each.

11. Insert the geometric means. Give all possible answers.
(a) three means between 2 and 162 (b) four means between 486 and 2 (c) one mean between 7 and 28

12. Each set of expressions gives three consecutive terms of a geometric sequence. Determine x and the terms.
(a) $x-4,\ x+2,\ x+14$ (b) $x,\ x+6,\ 4x$

13. In a geometric sequence with $r=2$, the third term is $y-1$ and the sixth term is $5y+22$. Determine y, then t_1 and t_8.

14. The general term of the sequence $5,-15,45,\ldots$ is
A. $t_n=5(3)^{n-1}$ **B.** $t_n=-3(5)^{n-1}$ **C.** $t_n=(-15)^{n-1}$ **D.** $t_n=5(-3)^{n-1}$

15. In a geometric sequence $t_1=6$ and $t_4=162$. The value of t_7 is
A. 318 **B.** 1458 **C.** 4374 **D.** 13 122

16. Numerical Response The number of terms in the sequence $2,6,18,\ldots,39\,366$ is _____.

17. Numerical Response The expressions $x+2$, $x+5$ and $x+14$ are the first three terms of a geometric sequence. The value of t_4 is _____.

Lesson 6: Geometric Growth and Decay

Assignment

1. State the growth factor in each situation.

(a) A province's population rises by 1.5% per year.

(b) A car depreciates by 17% per year.

(c) A salary increases by 3.2% per year.

(d) A magazine loses 4% of its subscribers each year.

(e) A ball rebounds to $\frac{5}{6}$ of its previous height.

(f) An object loses one-third of its temperature above room temperature each minute.

2. A pendulum swings through an arc of 90 cm on its first swing. Each swing after that is 8% shorter than the one before.

(a) State the growth factor. _____

(b) Determine the length of the arc on the 6th swing, to the nearest tenth of a centimetre.

3. A snowmobile is worth \$16 500 on January 1, 2025. It depreciates by 13% per year. What is its value on January 1, 2031, to the nearest dollar?

4. A ball is dropped from a height of 25 m. Each bounce reaches 60% of the height of the bounce before.

(a) How high is the first bounce? _____

(b) Determine the height of the 5th bounce, to the nearest centimetre.

(c) Raj lets $t_1 = 25$ and says the 5th bounce is t_5. Explain his mistake.

5. A culture of 150 bacteria doubles every 30 minutes.

(a) How many bacteria are there after 3 hours?

(b) After how many whole doublings does the number first exceed 1 000 000? How many hours is that?

6. Iodine-131 has a half-life of 8 days. A lab has a 200 mg sample.
(a) How much is left after 40 days?

(b) After how many whole half-lives is there first less than 1 mg left? How many days is that?

7. (a) \$3200 is invested at 4.25% per year compounded annually. Determine its value after 6 years.

(b) An investment of \$4200 is worth \$6000 after 8 years, compounded annually. Determine the annual interest rate, to the nearest tenth of a percent.

8. The value of a coin collection is modelled by $t_n = 1800(1.065)^{n-1}$, where t_1 is its value today.
(a) By what percent does its value increase each year? _____
(b) Determine its value 10 years from now, to the nearest cent.

(c) By what percent, to the nearest whole percent, has its value grown over those 10 years?

9. \$1500 is invested at 5.5% per year, compounded annually. How many full years does it take for the investment to double?

10. A young tree grows 1.6 m in its first year. In each year after that, its growth is 80% of its growth the year before.
(a) How much does it grow in the 5th year, to the nearest centimetre?

(b) In which year does it first grow less than 0.25 m?

11. In a store's clearance sale, an item's price drops by 5% of the previous day's price each day it is not sold. A jacket costs \$480 on day 1 of the sale.
(a) State the growth factor. _____
(b) What is the price on day 8, to the nearest cent?

(c) Write an expression for the price on day n. _____
(d) The jacket sold for \$302.52. On which day was it sold?

12. A 4 cm by 6 cm photo is enlarged on a photocopier: each copy is 20% longer and 20% wider than the one it is copied from.
(a) State the growth factor for the length and width. _____

(b) Determine the dimensions after 3 enlargements.

(c) Determine the area of the original and of the first two enlargements. What is the growth factor for the area?

(d) Determine the area after 5 enlargements, to the nearest hundredth.

13. Marco deposits \$2000 at the start of each year into a GIC paying 4% per year, compounded annually. What is the total value of his GIC deposits just before he makes his 4th deposit?

14. In 2020 a house was worth \$380 000. Its value increases by 5% per year. Which equation gives its value V, t years after 2020?

A. $V = 380\,000(0.05)^t$ **B.** $V = 380\,000(1.05)^{t-1}$ **C.** $V = 380\,000(1.05)^t$ **D.** $V = 380\,000(0.95)^t$

15. Money is invested at 2.6% per year, compounded annually. The ratio of the amount at the end of any year to the amount at the start of that year is

A. 0.026 **B.** 1.026 **C.** 2.6 **D.** 0.974

Use the following information to answer the next two questions.
A new motor home costs \$95 000. It loses 18% of its value in the first year and 8% of its value each year after that.

16. The value of the motor home 5 years after it is bought, to the nearest dollar, is

A. \$35 220 **B.** \$51 342 **C.** \$55 807 **D.** \$62 613

17. Numerical Response The minimum number of whole years until the motor home is worth less than \$45 000 is _____.

18. Numerical Response A culture of 40 cells triples every hour. The number of cells after 5 hours is _____.

Lesson 7: Geometric Series

Assignment

1. Determine each sum. Give exact answers unless told otherwise.

(a) $2 + 10 + 50 + \cdots \quad (S_8)$

(b) $640 + 320 + 160 + \cdots \quad (S_{10})$

(c) $1 - 4 + 16 - \cdots \quad (S_7)$

(d) $\frac{1}{9} + \frac{1}{3} + 1 + \cdots \quad (S_6)$

(e) $50 + 60 + 72 + \cdots \quad (S_{12}$, nearest tenth)

(f) $-81 + 54 - 36 + \cdots \quad (S_8)$

2. Determine each sum.

(a) $5 + 10 + 20 + \cdots + 5120$

(b) $2187 + 729 + 243 + \cdots + 1$

(c) $-2 + 6 - 18 + \cdots - 1458$

(d) $0.2 + 0.6 + 1.8 + \cdots + 145.8$

3. A geometric sequence has general term $t_n = 5(-2)^{n-1}$.

(a) Write the first three terms. _____

(b) Determine the sum of the first 10 terms.

4. In a geometric sequence, $t_3 = 72$ and $r = 3$. Determine the sum of the first 8 terms.

5. A geometric series has $r = 3$ and $S_6 = 1820$. Determine t_1.

6. A carpenter cuts a board into 6 pieces whose lengths form a geometric sequence. The shortest piece is 3 cm and the longest is 96 cm. How long was the board?

7. (a) How many terms of $7 + 14 + 28 + \cdots$ add to 7161?

(b) How many terms of $-4 - 12 - 36 - \cdots$ add to -4372?

(c) The series $5 - 10 + 20 - \cdots$ has a sum of -425. Determine the last term and the number of terms.

8. (a) A geometric series has $t_1 = 2$ and $S_3 = 62$. Determine all possible values of r.

(b) A geometric series has $t_1 = 10$ and $S_5 = 3410$. Use a table to determine r $(r > 0)$ and then t_3.

9. The sum of the first n terms of a series is $S_n = 3(4^n - 1)$.
(a) Determine the first four terms.

(b) Determine t_9 in two different ways.

10. Jonah puts 5 cents in a jar on day 1, 10 cents on day 2, 20 cents on day 3, and keeps doubling the amount each day.
(a) How much does he put in on day 12?

(b) How much is in the jar after 12 days?

(c) On which day does the total in the jar first pass \$1000?

11. In a yeast culture, 40 new cells form in the first hour. The numbers of new cells formed each hour make a geometric sequence. In the first 5 hours, 4840 new cells form in total. Use a table to determine r, then find the number of new cells formed in the 5th hour.

12. In a school's phone tree, the principal calls r staff members, each of them calls r others, and so on. In 6 levels, 1092 people are called. Use a table to determine r.

13. A ball is dropped from 12 m. Each bounce reaches 75% of the height before.

(a) Determine the height of the 4th bounce, to the nearest centimetre.

(b) Determine the total vertical distance travelled when it hits the ground for the 5th time, to the nearest centimetre.

(c) How many times has it hit the ground when its total distance first passes 75 m?

14. Which expression gives the sum of the first 10 terms of $6 + 18 + 54 + \cdots$?

A. $\frac{6(3^{10} - 1)}{2}$ **B.** $\frac{6(3^{9} - 1)}{2}$ **C.** $\frac{6(3^{10} - 1)}{3}$ **D.** $6(3)^9$

15. For a series, $S_n = 4(2^n - 1)$. The value of t_6 is

A. 124 **B.** 128 **C.** 252 **D.** 256

16. A geometric series has $t_1 = 3$ and $r = 1$. The value of S_{20} is

A. 3 **B.** 20 **C.** 60 **D.** undefined

17. Numerical Response In a geometric sequence, $t_2 = 6$ and $t_5 = 162$. The sum of the first 6 terms is _____.

18. Numerical Response A ball is dropped from 4 m. Each bounce reaches $\frac{3}{4}$ of the height before. The total vertical distance it has travelled when it hits the ground for the 5th time, to the nearest tenth of a metre, is _____.

Lesson 8: Infinite Geometric Series

Assignment

1. Consider the infinite geometric series $10 + 8 + 6.4 + \ldots$

(a) Complete the table of partial sums.

n	1	2	3	4	5
S_n					

(b) Determine S_{20} to the nearest hundredth.

(c) Determine S_∞.

(d) Explain why S_{20} is so close to S_∞.

2. Write the first five partial sums of each infinite geometric series and explain why the series is divergent.

(a) $1 + 1.5 + 2.25 + \ldots$

(b) $3 - 3 + 3 - 3 + \ldots$

3. Determine r and state whether each series is convergent or divergent. Calculate S_∞ where it exists.

(a) $54 + 18 + 6 + \ldots$ (b) $7 - 3.5 + 1.75 - \ldots$ (c) $0.2 + 0.6 + 1.8 + \ldots$

(d) $6 + 6 + 6 + \ldots$ (e) $100 - 90 + 81 - \ldots$ (f) $-24 - 18 - 13.5 - \ldots$

(g) $\frac{5}{2} + \frac{5}{6} + \frac{5}{18} + \ldots$ (h) $8 - 8 + 8 - \ldots$ (i) $0.81 + 0.9 + 1 + \ldots$

4. Consider the series $200 + 160 + 128 + \ldots$

(a) Determine S_∞ and S_{10} (to the nearest hundredth).

(b) Use a table of values to find the smallest number of terms whose sum is greater than 990.

5. Each part describes an infinite geometric series. Determine the missing value.

(a) $S_\infty = 60$, $r = 0.35$; find t_1. (b) $S_\infty = -45$, $r = -0.5$; find t_1. (c) $S_\infty = 20$, $r = \frac{2}{3}$; find t_1.

(d) $t_1 = 9$, $S_\infty = 12$; find r. (e) $t_1 = 30$, $S_\infty = 20$; find r. (f) $t_1 = 14$, $S_\infty = 7$; find r.

6. In an infinite geometric series, $t_2 = 24$ and $t_4 = 6$. Determine t_1 and S_∞ for each of the two possible series.

7. The expressions $5x^2$, $5x$ and 5 are the first three terms of an infinite series ($x \neq 0$).

(a) Show that the series is geometric.

(b) Explain why the series has a sum to infinity when $x = 4$, and determine it.

(c) Explain why there is no sum to infinity when $x = \frac{1}{2}$.

8. Use an infinite geometric series to write each repeating decimal as a fraction in lowest terms.

(a) $0.\overline{8}$ (b) $0.\overline{45}$ (c) $0.\overline{126}$

(d) $2.\overline{3}$ (e) $0.4\overline{6}$ (f) $0.1\overline{36}$

(g) Use a series to show that $0.\overline{9} = 1$.

9. A ball is dropped from a height of 4 m. After each bounce it rises to $\frac{3}{4}$ of the height it fell from.
(a) If the ball bounces forever, what total vertical distance does it travel?

(b) How far has it travelled, to the nearest hundredth of a metre, at the moment it hits the ground for the sixth time?

10. Each morning a patient takes a 200 mg dose of a medicine. By the next morning, 25% of the medicine in the body is still there. Just after the nth dose, the amount in the body is $200 + 200(0.25) + 200(0.25)^2 + \ldots$ (n terms).
(a) How much medicine is in the body just after the third dose?

(b) If the patient keeps taking the medicine for a long time, what amount (to the nearest hundredth of a milligram) does the level just after a dose approach?

11. A gold mine produced 3000 kg of gold in its first year. Each year after that, its production was 12% less than the year before.
(a) How much gold, to the nearest kilogram, did it produce in year 6?

(b) How much gold, to the nearest kilogram, did it produce in the first 10 years?

(c) What is the most gold the mine could ever produce if this pattern continued?

(d) Give one reason the actual total could be less than your answer to (c).

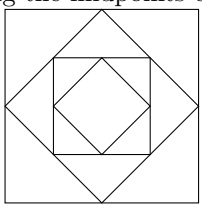
12. Consider the infinite geometric series $1 + (2x - 1) + (2x - 1)^2 + \ldots$

(a) For what values of x does the series converge?

(b) Determine S_∞ when $x = 0.75$.

(c) Determine x if $S_\infty = \frac{5}{4}$.

13. A square has sides of 8 cm. Joining the midpoints of its sides makes a second square with half the area of the first. Joining the midpoints of the second square makes a third square, and so on forever.



(a) Write the areas of the first four squares. _____

(b) Determine the total area of all the squares.

14. Which infinite geometric series is convergent?

A. $2 + 3 + 4.5 + \ldots$ **B.** $5 - 5 + 5 - \ldots$ **C.** $12 - 9 + 6.75 - \ldots$ **D.** $0.1 + 0.2 + 0.4 + \ldots$

15. The sum to infinity of a geometric series is 48 and its first term is 36. The common ratio is

A. $\frac{1}{4}$ **B.** $\frac{3}{4}$ **C.** $\frac{4}{3}$ **D.** $-\frac{1}{4}$

16. For $x > 1$, an expression for the sum to infinity of $\frac{1}{x} + \frac{1}{x^2} + \frac{1}{x^3} + \ldots$ is

A. $\frac{1}{x-1}$ **B.** $\frac{x}{x-1}$ **C.** $\frac{1}{1-x}$ **D.** $x - 1$

17. Numerical Response An infinite geometric series has $r = 0.4$ and $S_\infty = 75$. The value of t_3, to the nearest tenth, is _____.

18. Numerical Response When $0.\overline{24}$ is written as a fraction $\frac{a}{b}$ in lowest terms, the value of $a + b$ is _____.

Lesson 9: Unit 1 Review

Assignment

Calculators are allowed. Round as each question asks.

Multiple Choice

1. Which sequence is neither arithmetic nor geometric?

A. $4, 9, 14, 19, \ldots$ **B.** $4, 12, 36, 108, \ldots$ **C.** $4, 6, 9, 13.5, \ldots$ **D.** $4, 7, 12, 19, \ldots$

2. In the arithmetic sequence $7, 12, 17, \ldots$, the number 222 is the term

A. t_{43} **B.** t_{44} **C.** t_{45} **D.** t_{46}

3. In a geometric sequence, $t_1 = 5$ and $t_4 = -40$. The value of t_7 is

A. -640 **B.** -320 **C.** 320 **D.** 640

4. The sum of the first n terms of a sequence is $S_n = 3n^2 - n$. The general term is

A. $t_n = 6n - 4$ **B.** $t_n = 6n - 2$ **C.** $t_n = 6n + 4$ **D.** $t_n = 3n^2 - n$

5. The points (n, t_n) of an arithmetic sequence lie on a line with slope -3 that passes through $(5, 4)$. The first term is

A. -8 **B.** 1 **C.** 16 **D.** 19

6. A car bought for \$28 500 loses 18% of its value each year. To the nearest cent, its value 4 years after it was bought is

A. \$7980.00

B. \$10 566.09

C. \$12 885.47

D. \$15 713.99

7. An infinite geometric series has $S_\infty = 25$ and $r = -0.6$. The first term is

A. -15 **B.** 10 **C.** 15.625 **D.** 40

8. Three arithmetic means are inserted between 5 and 33. The common difference is

A. 5.6 **B.** 7 **C.** $9.\overline{3}$ **D.** 14

9. The sum of the geometric series $3 + 6 + 12 + \cdots + 768$ is

A. 765 **B.** 1533 **C.** 1536 **D.** 3069

Numerical Response

10. Numerical Response The number of terms in the arithmetic sequence $15, 22, 29, \ldots, 701$ is _____.

11. Numerical Response In a geometric sequence, $t_3 = 20$ and $t_6 = 67.5$. The first term, to the nearest tenth, is _____.

12. Numerical Response Cans are stacked in rows. The bottom row has 30 cans, each row above has 2 fewer cans than the row below it, and the top row has 4 cans. The total number of cans is _____.

13. Numerical Response When $0.1\overline{5}$ is written as a fraction $\frac{a}{b}$ in lowest terms, the value of b is _____.

Written Response

14. A sequence has general term $t_n = 25 - 3n$.

(a) Write the first five terms. Is the sequence arithmetic or geometric? Is its graph linear?

(b) Which term is -50?

(c) Determine the first negative term.

15. (a) An arithmetic sequence has $t_1 = -8$ and $d = 3.5$. Determine t_{20} and the general term.

(b) Insert four arithmetic means between 10 and -5.

16. In an arithmetic sequence, $t_6 = 41$ and $t_{15} = 95$. Determine d, t_1, the general term and t_{40}.

17. \$4000 is invested at 3.25% per year simple interest (the interest each year is 3.25% of \$4000).

(a) Write the value of the investment at the end of years 1, 2 and 3.

(b) Write a general term for the value at the end of year n.

(c) At the end of which year is the value first more than \$6000?

18. (a) Determine the sum of the first 40 terms of $12 + 17 + 22 + \ldots$

(b) Determine the sum of all the multiples of 7 between 100 and 500.

19. How many terms of the arithmetic series $4 + 6 + 8 + \ldots$ must be added to get a sum of 340?

20. Insert three geometric means between 2 and 162. Give both possible answers.

21. A culture starts with 500 bacteria, and the number doubles every 3 hours.
(a) How many bacteria are there after 24 hours?

(b) After how many hours does the number first pass 1 000 000?

22. \$2500 is invested at 4.5% per year, compounded annually.
(a) Write the value at the end of years 1, 2 and 3, to the nearest cent.

(b) Write a general term for the value at the end of year n, and determine the value at the end of year 10.

23. (a) A geometric series has $t_1 = 6$ and $r = 1.5$. Determine S_8 to the nearest hundredth.

(b) Determine the number of terms and the sum of $1250 + 250 + 50 + \cdots + 0.08$.

24. A post is shared with 4 people in round 1. In each round after that, each person who just received it shares it with 4 new people.

(a) How many people in total have received the post after 7 rounds?

(b) In which round does the total number of people first pass one million?

25. (a) Determine the sum to infinity of $64 - 48 + 36 - \ldots$

(b) For what values of x does $1 + 2x + 4x^2 + \ldots$ converge?

(c) Determine x if the sum to infinity of the series in (b) is 5.

26. A hot-air balloon lifts off from the ground. It rises 300 m in the first minute, and in each minute after that it rises 85% as far as it rose in the minute before.

(a) How far, to the nearest hundredth of a metre, does it rise during the 4th minute?

(b) What is its height, to the nearest hundredth of a metre, after 10 minutes?

(c) During which minute does it first rise less than 30 m?

(d) What height does the balloon approach? What must you assume?